A Novel Thermal Uniformity Index for Honeybee Colony Stability Assessment Using Infrared Thermography and Fuzzy Logic Inference System

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***Abstract*— Infrared thermography offers a non-invasive method of measurement of the thermal activity of colonies of honeybees. Relatively steady temperature distributions are realized in stable colonies within the cluster and can be examined with the help of thermal imaging data. This paper presents a Novel Thermal Uniformity Index (TUI) to determine the stability of the honeybee colonies with the use of infrared thermography and fuzzy logic classification. The analysis of the thermal images recorded with the use of an infrared camera (HIKMICRO) was performed to obtain temperature features of the colony area such as maximum, minimum, and average temperature. The thermography report stated an average colony temperature of 31 °C (ranging between 22.9 °C and 37.6 °C) and the colony had an active thermoregulation. The thermal stability in the proposed TUI is measured by studying the temperature distribution in the cluster region. They are then classified into one of the three linguistic states of stable, moderate and disturbed using a fuzzy inference system (FIS). The regularity of thermal behavior of the colony over the observation period is exhibited by temporal temperature variation obtained by the temperature curve report. As it has been experimentally verified, the proposed approach is efficient in interpreting the patterns of thermographic data and can be used as a simple and yet effective method of monitoring the stability of the colony. The presented framework presents an effective research instrument of non-contact colony health diagnostic in precision apiculture.**

Keywords—Infrared Thermography, Honeybee Colony Monitoring, Thermal Uniformity Index, Fuzzy Inference System, Precision Apiculture.

I. Introduction

Honeybee colonies are a complicated biological system which ensures the internal stability by the collective behavior and environmental control. Thermoregulation is one of the most significant colony survival characteristics with bees controlling the best temperatures in the hive to ensure the brood and colony activity. Tracking of the state of colonies with the help of modern sensing methods that include thermal imaging, acoustic sensing and intelligent monitoring systems has become a topic of great interest in recent years. The technologies allow non-invasive studying of the colony health, response to environmental conditions, and behavior dynamics, which can be useful in the precision apiculture and economical practices of beekeeping. The use of temperature sensing in hives has been used as a quantitative measure of the strength of the colony. Statistical correlation techniques combined with distributed temperature sensing were used to establish relationships between internal hive temperature variations, brood density, and colony population dynamic. Constant thermos observations have shown that steady temperature regimes are associated with intense brood activity and productiveness of colonies and have been used to back temperature-based non-invasive monitoring systems of colony health **[1]**. Combining thermal imaging with deep learning algorithms will allow the accurate classification and recognition of honeybee's castes. Infrared (IR) thermal images carry differentiating spatial features that identify the workers, drones, and queens. These signatures are used in advanced computation models to provide high accuracy in classification and hence greatly improve automation and accuracy in biological and behavioral analyses of honeybee colonies **[2]**. IR thermography leads to the discovery of the thermal ecology of bees by defining their temperature control systems. The obtained thermal signatures offer a peephole in the physiological reaction under foraging and environment adaption guiding the interpretation of ecological interaction and behavioral pattern. The methodologies are useful in investigating the pollinator reactions to environmental variability **[3]**. Monitoring of the temperature of hives in longer periods helps in determining the best thermal comfort zones needed to develop the colony. Consistent thermal conditions are very important in brood development and productivity of colonies in general. Nonstop environmental data analytics help optimize the perception of internal hive processes, which assist in developing smarter and dynamic monitoring frameworks **[4]**. The acoustic signal analysis of beehives provides the data about the activity of the colony, stress, and change of the environment. Differences between acoustic patterns indicate behavioral as well as physiological condition of the colony. This method being a non-invasive method in monitoring offers real-time diagnostics and when used in combination with other sensing modalities, it increases the effectiveness of the entire system **[5]**. The measurement of physiological and environmental markers allows making a comprehensive evaluation of the health and nutrition of the colony. The accuracy of identifying the stability of the colony is enhanced by the combination of various parameters. This combined approach increases the understanding of environmental changes on the population of bees and leads to sustainable practices in beekeeping **[6]**. Integration of infrared thermography and optimization processes enhances the hidden thermal anomalies. Complex computational methods are used to sharpen the characterization and definition of thermographic data. Such techniques make the detection of minor defects and changes more accurate, proving the increased universality of thermographic analysis in complicated systems **[7]**. The combination of thermal imaging

and data mining algorithms can be used to predict honey and beeswax harvest. Predictive accuracy is improved meaningfully by the use of environmental and thermal data. These models aid the use of data to make decisions in the operations of beekeeping and highlight the significance of precision apiculture by using advanced sensing technologies **[8]**. Spatial modeling methods are used to examine temperature field and thermal processes in intricate settings. Knowledge of spatial heat variations increases the interpretation of the environmental condition. The methods can be used on biological systems like beehives to diversify the study of thermal behaviour on the natural and built environments **[9]**. Thermal cameras combined with optical cameras can improve precision in surveilling the activity of beehives. Machine learning methods can be used to interpret visual and thermal data in a complex manner. The multimodal sensing can give us a full picture of the colony behavior, which enhances real-time observation and analysis of the apiculture technologies **[10]**. The interactive and observational features of honeybee colonies are offered by robotic systems with sensing technologies. The systems allow uninterrupted observation of behavioral and environmental reactions improving efficiency and minimizing manual collection of data. These innovations create fresh possibilities of trial and application researches in apiculture **[11]**. Quantitative evaluation of colony development and productivity be done through quantitative monitoring techniques are very accurate. Organized data is collected to monitor the growth of colonies over a specific duration of time, which will be more reliable in evaluating performance and monitoring health. This is a method that helps in management and optimization strategies used in the keeping of bees **[12]**. The developed sensor technologies allow non-invasive monitoring of bee colonies with the help of various sources of data. The temperature, visual and acoustic indicators provide detailed diagnostic data. Combining different sensing modalities produces an increase in accuracy and reliability in monitoring, and supports the creation of intelligent and automated solutions of monitoring beehives **[13]**. There are behavioral and physiological adaptations of honeybees to cope with heat stress at both individual and colony levels. The monitoring structures of temperature play a major role in stabilizing the colony in response to changing environmental conditions. Such responses serve as a reminder of bees and their tolerance to thermal stress, and their adaptations to such responses are useful in the context of climate change due to better colony management **[14]**. Queen rearing and selection procedures focus on the measured environment and genetic control. Healthy and productive colonies are maintained through proper breeding practices. These strategies maintain uniformity and quality in apicultural systems thus leading to long term sustainability and stability of the colony **[15].** Although these have been developed, the majority of current research is centered on the principles of direct temperature measurement, behavioral observation, or sensor observation of the hives, and almost no one has concentrated on quantitative thermographic analysis of the temperature distribution within a cluster of bees. Thermographic measurements typically depend on such straightforward guidelines as maximum and minimum temperatures, which might not be capable of entirely describing the spatial permanence of thermal structures within the cluster of colonies. Additionally, most monitoring methods involve the use of sensors installed in the hive interfering with the natural behavior of the colony. IR thermography is a non-contact and quick visualization method that is able to identify fine temperature distributions on the colony cluster solution. Nevertheless, existing thermographic measurements do not have a uniform quantitative measure that can indicate the general thermal homogeneity and stability of the colony cluster. Thus, in order to enhance non-invasive colony monitoring systems, it is necessary to develop a dependable thermographic index that will be able to decipher the pattern of temperature distribution. In order to overcome this drawback, this work comes up with a TUI of assessing the stability of honeybee colonies through infrared thermography. The given index measures the homogeneity of temperature distributions in the colony cluster area with statistical temperature parameters being obtained with the help of thermographic photographs. A FIS is also incorporated to define the computed index to classify the conditions of the colonies into three linguistic categories namely stable, moderate and disturbed. The proposed framework brings together thermographic data and fuzzy-based decision analysis in order to give a straightforward, understandable, and non-invasive way of keeping track of the stability of honeybee colonies. The main objective of this work is:

- To study the distribution of temperatures of the honeybee colony clusters through infrared thermography.
- To obtain valuable thermal data like T_{max}, T_{min}, and T_{mean} of thermographic images.
- To design a novel TUI to measure thermal stability in the colony cluster.
- To use the FIS in order to interpret the calculated index and categorize the colony conditions.
- To show that infrared thermography could provide useful non-destructive method of measuring the stability of honeybee colonies and thermoregulation behavior.

Overall, the proposed work aims to improve the non-contact monitoring of honeybee colonies by providing a the quantitative thermographic indicator for colony stability assessment.

II. Proposed Architecture

The suggested architecture is designed as a serial processing setup of testing the stability of honeybee colonies based on infrared thermography and thermal analysis with fuzzy logic. The general architecture of the system is presented in **Fig. 1**. First, a thermal image of the honeybee colony cluster is taken by use of an IR thermal camera in a controlled environment. These thermographic images get the temperature distribution of the bee cluster on the surface, and provide valuable data on the thermoregulatory behaviour of the colony. The captured images are then subjected to thermal image-processing software to derive the temperature data of the cluster area. Image preprocessing is then carried out in the next step in order to improve the thermographic data. This will include segmentation of the colony cluster region, temperature calibration and eliminating the background noise to achieve accurate temperature measurements. Upon preprocessing, critical thermal parameters are obtained in the images including Tmax, Tmin and Tmean. These parameters define the spatial temperature distribution in the colony cluster. The obtained thermal parameters are then utilized to calculate the

proposed TUI which measures the stability of the temperature distribution in the colony cluster region. The resultant index is then fed as input to the FIS. The fuzzy system involves the application of predetermined membership and rule-based reasoning to understand the index values and categorize the colony conditions in three linguistic terms, namely, stable, moderate, and disturbed. Lastly the FIS findings are reviewed to assess the thermal stability of the honeybee colony and can be used to monitor the health of the colony and thermoregulatory behaviour in the infrared thermography without the need to interfere with the colony.

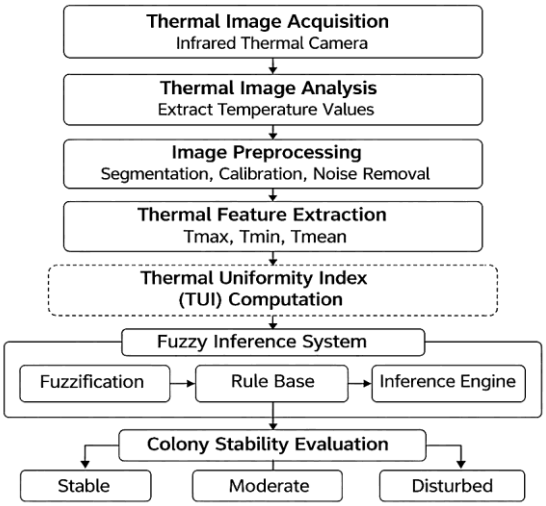

Fig. 1. Proposed thermography-based framework for honeybee colony stability analysis using thermal imaging and fuzzy inference system.

III. Data and Methods

This part presents the methodology of the suggested system of measuring the stability of honeybee colonies using infrared thermography and fuzzy inference analysis. The structure consists of a series of consecutive steps, such as thermal image capture, preprocessing, thermal feature, TUI calculation, and fuzzy-logic-based classification. This systematic method allows deploying an appropriate analysis of thermographic data to monitor the thermoregulation and stability of honeybee colonies non-invasively.

A. Data Acquisition

HIKMICRO infrared thermographic camera was utilized to collect thermal data to be used in this research as shown in **Fig. 2**. Thermal images of a cluster of honeybee's colonies were taken by the camera in the field observation conditions. The camera was placed around 0.90 m away of the bee cluster in the process of acquisition and the emissivity was set at 0.97 in order to measure accurately the temperature of the surface of the biological material. The temperature of the environment during the observation period was between 29.6 -36.7 °C, which indicated the ambient temperature around the given experiment. In the monitoring session, 17 thermographic images were taken to investigate the time and space change in temperature distribution of the colony cluster. This dataset showed that the highest recorded temperature was of around 37.6 °C and a lowest of about 22.9 °C. The temperature of the core of the bee cluster was nearly 32.2 °C, and the temperature of the cluster was in the range of 31.0 °C. Such measurements reflect that the colony has active thermoregulation and bees are united to maintain a constant thermal environment. The thermographic captured images give detailed maps of temperature distributions across the colony, thus allowing analysis of thermal gradient and pattern of heat concentration in the colony. These temperature regimes are indicators of the total metabolic activity and thermoregulatory processes of honeybees that are critical to determine the stability and health of the colony. The temperature data obtained out of these thermographic images is the main input to be used in further preprocessing, thermal characteristics extraction and fuzzy inference processes.

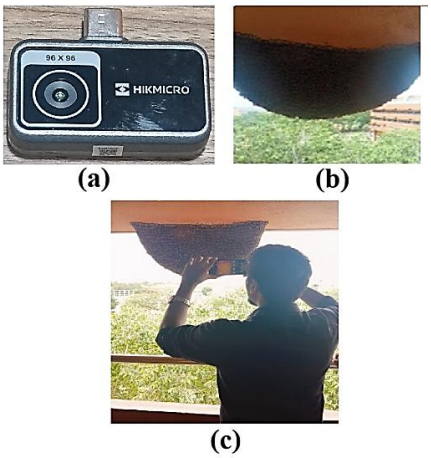

Fig. 2. Experimental setup for thermal monitoring of honeybee colonies: (a) portable HIKMICRO thermal imaging device used for capturing infrared data, (b) natural honeybee hive structure under field conditions at Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Chennai, India, and (c) real-time thermal data acquisition process using handheld imaging for non-invasive colony analysis.

B. Image Preprocessing

Thermographic images are first processed to improve the quality of data before thermal features are detected to obtain quality and reliable temperature data. Raw photographs might have background interference and noise of the environment that might interfere with temperature measurement. The preprocessing stage involves isolating the bee colony cluster from the background through segmentation, removing environmental and sensor noise, ensuring accurate temperature measurements from the infrared data, and applying contrast enhancement along with normalization to clearly highlight thermal variations within the colony. The operations performed during these preprocessing steps will ensure that the extracted temperature values are the truth about the actual thermal properties of the honeybee colony cluster.

C. Thermal Feature Extraction

After pre-processing, the thermographic image is converted into quantitative thermal variables that describe the temperature field in the honeybee cluster of colonies. Before performing the feature extraction, the bee colony area is separated from the thermal image to identify the region of interest. This is done using temperature thresholding, where the warmer cluster is distinguished from the cooler background. Then, simple image operations are applied to clean the region and improve the accuracy of the extracted temperature values. In contrast to other traditional image-analysis methods that rely on visuality, the work focuses on feature-extraction, which is thermographic and temperature-centric. Among the metrics of each thermal image, the following ones are obtained:

- Tmax – The highest temperature measured in the colony cluster.

- Tmin – The lowest temperature at the colony cluster.
- Tmean – The average temperature of the group of colonies.

According to the thermographic data, the average temperature of the colony cluster has a range of 22.9 °C to 37.6 °C with the core area having a temperature of around 32.2 °C, which is indicative of active thermoregulatory behaviour. These parameters outline the distribution pattern of heat in the colony cluster, hence indicate the overall behaviour of honeybees in producing heat.

D. *Thermal Uniformity Index (TUI)*

The Thermal Uniformity Index (TUI) is used to measure the thermal uniformity of the material within a heating system. A quantitative measure of thermal stability of a colony is presented, the Thermal Uniformity Index (TUI). The index evaluates the homogeneity of temperature distribution over the colony cluster area, and is given by **Equation (1)**.

$$TUI = 1 - \frac{\sigma_T}{T_{mean}} \tag{1}$$

where, T_{mean} represents the mean temperature of the colony cluster, σ_T represents the standard deviation of temperature distribution. An increase in TUI value indicates homogenous and stable thermal distribution of the colony cluster, but lower values indicate the disordered thermal behaviour, which could be a sign of colony stress or disturbance.

E. *Fuzzy Logic Inference System (FIS)*

In order to complement the interpretation of the thermal features, a fuzzy logic structure **[16]** is incorporated into the proposed methodology. This analysis can be performed with the fuzzy logic approach due to the fact that the conditions of stability of the colony evolve over time and cannot be sufficiently described with the help of strict numerical values. The FIS takes the TUI as its input. Three linguistic variables are established as low, moderate, and high stability with each of them having triangular membership functions characterized by **Equation (2)**.

$$\mu\,(x) = \begin{cases} 0, & x \le a \\ \frac{x-a}{b-1}, & a < x \le b \\ \frac{c-x}{c-b}, & b < x < c \\ 0, & x \ge c \end{cases} \tag{2}$$

where, x represents the input TUI value, a, b, c are the parameters defining the membership function, and $\mu(x)$= degree of membership between 0 and 1. The fuzzy rule base is constructed to classify colony stability based on TUI values. The fuzzy rules include:

- If TUI is High then the Colony is Stable.
- If TUI is Medium then the Colony is Moderately Stable,
- If TUI is Low then the Colony is Disturbed.

These rules are then processed by fuzzy inference engine and an ultimate decision is generated through defuzzification. The fuzzy membership functions are defined using triangular shapes due to their computational simplicity and efficiency. Parameters are selected based on the observed TUI range (0.89–0.95) and grouped into low, medium, and high stability levels. These functions are tuned to ensure smooth transitions and consistency with thermal behavior. The use of fuzzy logic is because honeybee colony thermal behavior is non-discrete and gradual and thus strict classification fails. It makes it possible to map the TUI in the linguistic states like low, medium, and high stability. It enhances flexibility and is more accurate to continuous patterns of thermoregulation.

F. *Thermal Pattern Analysis*

After extraction of features and processing of fuzzy inferences, the thermal parameters are explored so that patterns of temperature distribution in the honeybee colony cluster can be analyzed. Experimental measurements have shown that the distribution of thermal values of stable colonies at the cluster center is quite consistent with a temperature of about 31 °C, whereas the temperature values vary moderately at the edges of the colony between 23 °C and 37 °C. This thermal gradient implies the thermoregulation process of honeybee colonies in nature.

G. *Colony Stability Assessment*

The last phase of the proposed structure will assess the colony stability in terms of calculated results of Thermal Uniformity Index and fuzzy inference. Stable colonies are those with greater values of TUI and uniform patterns of temperature distribution. On the other hand, the colonies that have irregular thermal patterns and have low index values can be the one that have moderate or disturbed conditions. Through a combination of infrared thermography, the proposed Thermal Uniformity Index, fuzzy inference analysis, a non-invasive monitoring method of determining honeybee colony thermoregulation and stability is obtained.

IV. Result Analysis

The thermographic experiment has been done on a colony cluster of honeybees using a handheld HIKMICRO infrared camera under controlled observation conditions and emissivity was set at 0.97, camera distance was 0.90m, ambient temperature was 29.6 °C - 36.7 °C. The monitoring process was required to capture 17 thermographic images to examine the spatial and temporal distribution of temperature in the colony cluster. **Table 1** summarized the calculated values of the thermal parameters, TUI and the estimated standard deviation and indicated the association between the temperature change and thermal stability of the cluster of honeybee colony. The thermographic data results show that the highest temperature was around 37.6 °C and the lowest temperature was around 22.9 °C. The core area of the honeybee cluster always recorded a temperature to be approximately 32.2 °C, but the average temperature of the whole colony was near 31 °C, hence proving the thermoregulation behaviour to be stable in the colony. Such findings suggest that honeybees coordinate heat regulation to ensure that the colony has a favorable thermal environment regarding its survival and functioning. To be clear and to make a valuable visualization, representative thermographic images are used only in this study. The chosen photos presented in **Fig. 3- Fig. 8** depict the patterns of distributions of temperatures throughout the colony cluster. These values

highlight the difference in temperature in the center and periphery of the colony cluster. Also, the temporal change of the parameters of temperature is shown by the temperature curve in **Fig. 9** which illustrates how the maximum, minimum and center temperatures vary over time. The findings are also clear demonstrating that the honeybee colony has fairly constant central temperature with regulated variation in outer areas which supports the efficiency of collective thermoregulation behaviour.

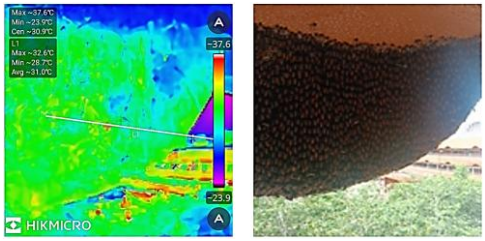

Fig. 3. Thermographic image representing maximum temperature distribution within the honeybee colony cluster (Tmax ≈ 37.6 ° C).

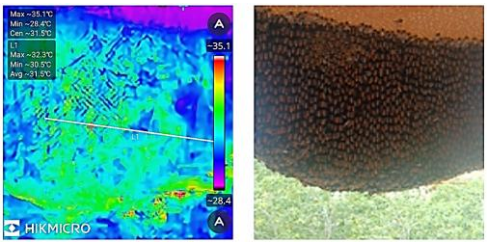

Fig. 4. Representative thermographic image showing typical temperature distribution pattern within the honeybee colony cluster.

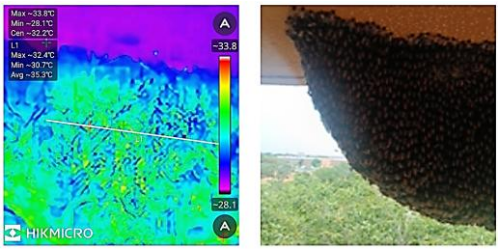

Fig. 5. Thermographic image illustrating spatial variation of temperature across the honeybee colony cluster.

The resultant thermal parameters and their distribution patterns can be trusted upon to compute the suggested TUI and additionally consider additional evaluation of the stability of the colony through the analysis of fuzzy inferences. The calculated values of TUI, shown in **Fig.10**, reflect that the colony possesses a fairly good thermal uniformity and thus the thermoregulation behaviour is stable throughout the samples that are observed. The processing time of thow work is low and suitable for near real-time applications. Preprocessing, and feature extraction are pixel-wise operations with a time complexity of O(N), where Nis the number of pixels. The TUI computation and fuzzy inference are O(1), resulting in efficient processing within a few seconds per image. Although the proposed work is simple, low-cost, and effective, its main limitation is the small dataset of 17 thermographic images, which may not capture long-term variations. Future work will focus on continuous monitoring and large-scale data collection, and implementing the advanced AI-based methods to improve colony health prediction and automated monitoring systems.

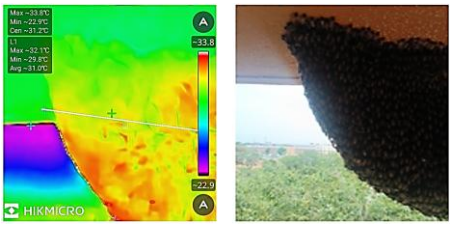

Fig. 6. Thermographic image representing minimum temperature distribution within the colony cluster (Tmin ≈ 22.9 ° C)

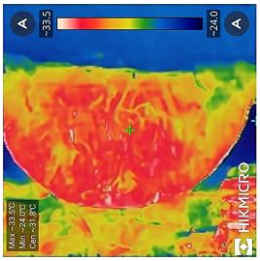

Fig. 7. Thermographic image showing uniform temperature distribution indicating stable thermoregulation within the honeybee colony cluster.

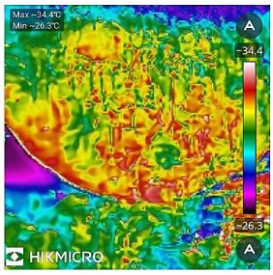

Fig. 8. Additional thermographic sample demonstrating consistent thermal behavior of the honeybee colony cluster.

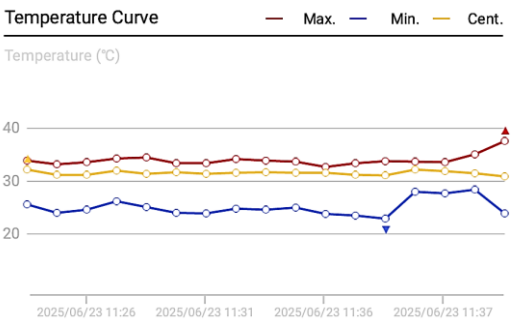

Fig. 9. Temperature variation curve showing maximum, minimum, and center temperature trends over time for the honeybee colony cluster.

TABLE I. EXTRACTED THERMAL FEATURES AND COMPUTED THERMAL UNIFORMITY INDEX (TUI)

Sample	Tmax (°C)	Tmin (°C)	Tmean (°C)	σT (approx.)	TUI
S1	37.6	23.9	31.0	3.425	0.890
S2	35.1	28.4	31.4	1.675	0.947
S3	33.6	27.7	31.6	1.475	0.953
S4	33.7	28.0	31.4	1.425	0.955
S5	33.8	22.9	31.0	2.725	0.912

Sample	Tmax (°C)	Tmin (°C)	Tmean (°C)	σT (approx.)	TUI
S1	37.6	23.9	31.0	3.425	0.890
S6	33.4	23.5	31.1	2.475	0.920
S7	32.7	23.8	31.1	2.225	0.928
S8	33.7	25.0	31.3	2.175	0.931
S9	33.9	24.6	31.2	2.325	0.925
S10	34.2	24.8	31.4	2.350	0.925
S11	33.4	23.9	31.1	2.375	0.924
S12	33.4	24.0	31.2	2.350	0.925
S13	34.5	25.1	31.5	2.350	0.925
S14	34.3	26.2	31.6	2.025	0.936
S15	33.6	24.6	31.2	2.250	0.928
S16	33.2	24.0	31.1	2.300	0.926
S17	33.9	25.6	31.3	2.075	0.934

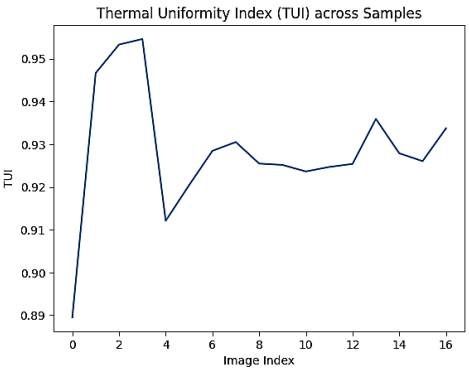

Fig. 10. Thermal Uniformity Index (TUI) across samples.

V. Conclusion

This current study was aimed at reviewing the use of infrared thermography along with a TUI and a FIS in the non-invasive measurement of the stability of honeybee colonies. The thermal distributions of honeybee colony were analyzed by obtaining a thermal image of the colony cluster in terms of infrared thermography camera and deriving key temperature parameters in terms of Tmax, Tmin and Tmean. The experimental findings showed that the honeybee colony is thermoregulated in the sense that it has a constant temperature value of about 32°C and has a uniform thermal distribution. On the other hand, any deviations in the temperature of the peripheral regions are indicative of the interactions with the surrounding and boundary conditions in the clusters. The suggested TUI was able to measure the consistency of the temperature distribution in the cluster of colonies. Additionally, the FIS was able to interpret the calculated index, and the states of stability were differentiated in terms of the condition of the colonies. The results of the research indicate that infrared thermography, TUI and fuzzy reasoning can provide an easy, effective, and non-contact approach to monitoring the stability of honeybee's colonies. The approach has great potential as it can be used in precision apiculture and smart hive observatory. Future studies can be carried out by adding to the dataset, adding time-series analysis to the dataset, and incorporating the more advanced machines - learning methods to improve colony health prediction and automated monitoring systems.

Acknowledgment

The author expresses sincere gratitude to Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Chennai, India, for providing the HIKMICRO infrared camera, honeybee colony images from the field site, and the laboratory facilities required to carry out this research

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